SUPPLEMENTARY MATERIALS – S3

Sensitivity Analysis

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S3.1 OBJECTIVE OF THE SENSITIVITY ANALYSIS

The sensitivity analysis addresses two questions.
First: How robust is the MROI conclusion against uncertainty in key parameters?
Second: Which parameters most strongly influence the MROI value and therefore require priority attention during empirical validation?

Three parameters are examined systematically:

- Regional grid carbon intensity (grid_CO_2)
- JIM-30 score
- Infrastructure Debt Multiplier (D)

Each parameter is varied individually while holding the others constant (one-at-a-time analysis), followed by a combined scenario analysis.

S3.2 BASE PARAMETERS (REFERENCE CASE)

Parameter	Solar Archetype	Wind-Storage Archetype
ΔE_saved (kWh/year)	10,000	45,000
grid_CO_2 (kg/kWh)	0.35	0.35
RACF (kg CO_2/capita/year)	526	526
JIM-30 (%)	25%	92%
D multiplier	0.3	0.3
EFU_input (direct)	6.65	29.94
EFU_input (corrected)	8.15	31.02
MROI (baseline)	**–3.0%**	**+27.1%**

S3.3 VARIABLE 1 – GRID CARBON INTENSITY (grid_CO_2 ±20%)

The Hungarian electricity grid carbon intensity fluctuates with changes in the energy mix. The MAVIR 2024–2025 annual average is 0.35 kg/kWh. With increased renewable penetration

this may decrease (optimistic scenario: 0.28 kg/kWh), while peak periods may yield higher values (pessimistic scenario: 0.42 kg/kWh).

S3.3.1 Solar Archetype – grid_CO_2 Sensitivity

grid_CO_2 (kg/kWh)	CO_2 savings (kg/year)	RACF units	EFU_input (corrected)	MROI	Classification
0.28 (–20%)	2,800	5.32	8.15	–3.7%	PARASITIC
0.315 (–10%)	3,150	5.99	8.15	–3.4%	PARASITIC
0.35 (baseline)	**3,500**	**6.65**	**8.15**	**–3.0%**	**PARASITIC**
0.385 (+10%)	3,850	7.32	8.15	–2.6%	PARASITIC
0.42 (+20%)	4,200	7.98	8.15	–2.1%	PARASITIC

Finding:
The solar archetype MROI remains negative across the full ±20% grid_CO_2 variation range. The conclusion is robust.

S3.3.2 Wind-Storage Archetype – grid_CO_2 Sensitivity

grid_CO_2 (kg/kWh)	CO_2 savings (kg/year)	RACF units	EFU_input (corrected)	MROI	Classification
0.28 (–20%)	12,600	23.95	31.02	+22.8%	STABLE
0.315 (–10%)	14,175	26.95	31.02	+24.9%	STABLE
0.35 (baseline)	**15,750**	**29.94**	**31.02**	**+27.1%**	**SYMBIOTIC**
0.385 (+10%)	17,325	32.94	31.02	+29.2%	SYMBIOTIC
0.42 (+20%)	18,900	35.93	31.02	+31.4%	SYMBIOTIC

Finding:
The wind-storage archetype remains positive across the entire grid_CO_2 range. Under –20% (high renewable penetration), the MROI declines into the STABLE band but does not become negative. The conclusion remains robust.

S3.4 VARIABLE 2 – JIM-30 SCORE (±15 percentage points)

Uncertainty in the JIM-30 score arises from subjective elements of component-level repairability assessment. The ±15 percentage-point variation reflects typical inter-rater reliability ranges.

S3.4.1 Solar Archetype – JIM-30 Sensitivity

JIM-30 (%)	Correction Multiplier	EFU_input (corrected)	MROI	Classification
10% (–15pp)	×1.270	8.45	–3.4%	PARASITIC
17.5% (–7.5pp)	×1.248	8.30	–3.2%	PARASITIC
25% (baseline)	**×1.225**	**8.15**	**–3.0%**	**PARASITIC**
32.5% (+7.5pp)	×1.203	8.00	–2.8%	PARASITIC
40% (+15pp)	×1.180	7.85	–2.6%	PARASITIC

Finding:
The solar archetype remains negative across the entire ±15pp range.

Critical threshold:
A JIM-30 value of 94% would be required to reach MROI = 0%, which is physically unattainable given sealed control electronics and server dependency.

S3.4.2 Wind-Storage Archetype – JIM-30 Sensitivity

JIM-30 (%)	Correction Multiplier	EFU_input (corrected)	MROI	Classification
77% (–15pp)	×1.069	32.01	+25.4%	SYMBIOTIC
84.5% (–7.5pp)	×1.047	31.34	+26.2%	SYMBIOTIC
92% (baseline)	**×1.024**	**31.02**	**+27.1%**	**SYMBIOTIC**
99.5% (+7.5pp)	×1.002	30.00	+27.9%	SYMBIOTIC
100% (+8pp cap)	×1.000	29.94	+28.0%	SYMBIOTIC

Finding:
The wind-storage archetype remains SYMBIOTIC across the entire ±15pp range.

S3.5 VARIABLE 3 – INFRASTRUCTURE DEBT MULTIPLIER D (0.1–0.5)

D Value	Solar MROI	Solar Classification	Wind MROI	Wind Classification
0.10	–1.2%	PARASITIC	+27.8%	SYMBIOTIC
0.15	–1.7%	PARASITIC	+27.6%	SYMBIOTIC
0.20	–2.1%	PARASITIC	+27.5%	SYMBIOTIC
0.25	–2.6%	PARASITIC	+27.3%	SYMBIOTIC
0.30 (baseline)	**–3.0%**	**PARASITIC**	**+27.1%**	**SYMBIOTIC**
0.35	–3.4%	PARASITIC	+26.9%	SYMBIOTIC
0.40	–3.9%	PARASITIC	+26.8%	SYMBIOTIC
0.45	–4.3%	PARASITIC	+26.6%	SYMBIOTIC

D Value	Solar MROI	Solar Classification	Wind MROI	Wind Classification
0.50	–4.8%	PARASITIC	+26.4%	SYMBIOTIC

Finding:
Across the full plausible D range (0.1–0.5), conclusions remain unchanged. The method is D-robust.

S3.6 COMBINED SCENARIO ANALYSIS

Scenario	grid_CO_2	JIM-30 Solar	JIM-30 Wind	D	Solar MROI	Wind MROI
Pessimistic (solar favorable, wind unfavorable)	0.42	40%	77%	0.10	–1.8%	+23.1%
Optimistic (solar unfavorable, wind favorable)	0.28	10%	100%	0.50	–5.1%	+28.0%
Baseline	0.35	25%	92%	0.30	–3.0%	+27.1%
Extreme pessimistic (all solar-favorable)	0.42	40%	77%	0.10	–1.8%	+23.1%

Finding:
Even under extreme combined assumptions, the solar archetype remains negative and the wind-storage archetype remains strongly positive.

S3.7 PARAMETER IMPORTANCE RANKING

Based on impact magnitude:

Solar Archetype

1. grid_CO_2
2. D
3. JIM-30

Wind-Storage Archetype

1. grid_CO_2
2. JIM-30
3. D

In both cases, grid_CO_2 is the dominant variable. However, no parameter variation changes the classification outcome.

S3.8 SUMMARY CONCLUSION

Three main findings emerge:

1. Robustness
The MROI classification (solar = PARASITIC, wind-storage = SYMBIOTIC) remains robust across the full plausible parameter ranges.

2. Threshold Distance
The solar archetype would require JIM-30 = 94% to reach MROI = 0%, which is physically unattainable given its structural characteristics.

3. Empirical Priority

grid_CO_2 is the most influential parameter for both archetypes. Integration of real-time MAVIR carbon intensity data in the 905.0 Pilot (2026–2027) is therefore the highest empirical refinement priority.

S3 Status:

Complete sensitivity analysis finalized. Individual and combined parameter variations confirm robustness of MROI conclusions.